THEORY BEHIND THE CALIBRATION: BREAK-EVEN SEPARABILITY AND THRESHOLD INSENSITIVITY

TEAM A1 — ROLE 1

Abstract. The calibration section states two facts it does not prove. First, that the break-even Bitcoin price is *additively separable*, with machine efficiency ε setting a power-bill floor and the hardware premium depending only on q/T and "not on ε at all… the two effects cancel exactly." Second, that the entry threshold $\bar{C}_M$ is insensitive to the outage penalty once $\beta + p \gg c + C_F$. Both are used to license headline numbers—the \$65k cheapest entry point, the \$43.6k power-bill floor, and a deliberately conservative $\beta + p = \$200$/MWh. We prove both, and verify that the proved formulas reproduce every cell of the calibration table.

1. Setup

A machine of efficiency ε (J/TH) earns, per MWh consumed, $R(\varepsilon) = \chi P_{\mathrm{BTC}}/\varepsilon$ with $\chi := b \cdot 3.6 \times 10^9/(t_B H)$ fixed by network conditions (b the block reward, t_B the block interval, H the network hash rate). Hardware at q \$/TH amortized over T years costs, per MW per hour,

$$C_M(\varepsilon, q, T) \;=\; \frac{q \cdot 10^6/\varepsilon}{8760\, T},$$

since one MW packs $10^6/\varepsilon$ TH/s. Entry occurs iff $C_M < \bar{C}_M = (R - p)\varphi^\circ$.

2. Break-even separability

Proposition 1 (The break-even price is additively separable, and the hardware premium does not depend on efficiency)**.** *Entry occurs iff $P_{\mathrm{BTC}} > P^*$, where*

$$\boxed{P^*(\varepsilon, q, T) \;=\; \underbrace{\frac{p}{\chi}\,\varepsilon}_{\textit{power-bill floor}} \;+\; \underbrace{\frac{10^6}{8760\,\varphi^\circ\,\chi} \cdot \frac{q}{T}}_{\textit{hardware premium}}}$$

The two terms are additively separable; the hardware premium depends on q/T alone and is exactly *independent of ε.*

Proof. Entry is $(R - p)\varphi^\circ > C_M$, i.e. $(\chi P_{\mathrm{BTC}}/\varepsilon - p)\varphi^\circ > q \cdot 10^6/(\varepsilon\, 8760\, T)$. Multiply both sides by $\varepsilon/\varphi^\circ > 0$:

$$\chi P_{\mathrm{BTC}} - p\,\varepsilon \;>\; \frac{q \cdot 10^6}{8760\,\varphi^\circ\, T},$$

and divide by $\chi > 0$. The ε appearing on the right has cancelled because *both* sides of the entry condition are proportional to $1/\varepsilon$: a more efficient machine earns proportionally more per MWh ($R \propto 1/\varepsilon$) and also packs proportionally more terahash into a MW, so it costs proportionally more per MW ($C_M \propto 1/\varepsilon$). The ratio that decides entry is efficiency-free; only the *power bill*, which is charged per MWh rather than per terahash, retains ε. □

Date: Supporting §7.

Corollary 2 (Efficiency competes down the floor; hardware markets set the premium)**.** *$\partial P^*/\partial\varepsilon = p/\chi > 0$ and $\partial P^*/\partial q > 0$, $\partial P^*/\partial T < 0$, with $\partial^2 P^*/\partial\varepsilon\,\partial(q/T) = 0$. Efficiency gains and hardware-price declines therefore lower the break-even price through* independent *channels, and a firm cannot trade one against the other.*

3. Numerical validation

Taking $p = \$50$/MWh and $\varphi^\circ = 145/155$, and calibrating χ from the reported efficiency coefficient, we recover $\chi = 0.019739$ and

$$\frac{p}{\chi} = 2533.0, \qquad \frac{10^6}{8760\,\varphi^\circ\,\chi} = 6181.9,$$

against the stated 2,533 and 6,183. Proposition 1 then reproduces the calibration table exactly:

	q (\$/TH)	$T=2$	$T=3$	$T=4$	$T=5$
Fleet average ($\varepsilon = 17.2$)	25	\$121k	\$95k	\$82k	\$74k
	35	\$152k	\$116k	\$98k	\$87k
	40	\$167k	\$126k	\$105k	\$93k
Frontier ($\varepsilon = 13.5$)	25	\$111k	\$86k	\$73k	**\$65k**
	35	\$142k	\$106k	\$88k	\$77k
	40	\$158k	\$117k	\$96k	\$84k

The $q \to 0$ limit gives the power-bill floor $2533 \times 17.2 = \$43.6$k at fleet-average efficiency, and the cheapest cell—frontier hardware, favourable prices, patient five-year amortization—is \$65k, both as stated.

4. Threshold insensitivity

Proposition 3 ($\bar{C}_M$ is insensitive to the outage penalty once $\beta + p \gg c + C_F$)**.** *With $\varphi^\circ = 1 - C_F/(\beta + p - c)$ and $\bar{C}_M = (R-p)\varphi^\circ$,*

$$\frac{\partial\,\bar{C}_M}{\partial(\beta+p)} = \frac{(R-p)\,C_F}{(\beta+p-c)^2} > 0, \qquad \bar{C}_M \uparrow (R-p) \quad \textit{as} \quad \beta + p \to \infty,$$

so the threshold is increasing but at a rate falling like $(\beta+p)^{-2}$, and it is bounded above by the miner margin $R-p$ regardless of how the outage penalty is set.

Proof. Differentiate φ° and multiply by $R-p$; the limit is $\varphi^\circ \to 1$. □

At the calibration ($R - p = 19$, $c = 45$, $C_F = 10$):

$\beta + p$	φ°	$\bar{C}_M$	$\partial\bar{C}_M/\partial(\beta+p)$
100	0.8182	15.545	0.0628
200	0.9355	17.774	0.0079
400	0.9718	18.465	0.0015
1000	0.9895	18.801	0.0002

At $\beta + p = \$200$/MWh a further \$100/MWh of assumed outage cost moves $\bar{C}_M$ by under \$0.80/MWh, against a threshold near \$18/MWh. Proposition 3 is what licenses the deliberately conservative choice of $\beta + p$: the entry margin is a fact about $R - p$, not about how the regulator values lost load.

Remark 4 (What this does and does not settle). These two propositions validate the calibration's *internal* logic—the functional form of P^* and the robustness of $\bar{C}_M$. They say nothing about the input values. In particular the tariff $p = \$50/\text{MWh}$ is an assumption, not a measurement; rebuilding it from utility-level data moves $\bar{C}_M = (R - p)\varphi^\circ$ one-for-one and is the single most consequential empirical check on the entry threshold.